Taxi Sale System

SRS File

Scope

The TaxiSales System is a full-stack web application designed to digitize and manage transport fleet operations. The core capabilities of the system include:

Secure user authentication and role-based access control.

Comprehensive registration and profile management for partner transport companies (operators).

Real-time tracking of taxi trips, invoices, payment types, and sales transaction headers (tblsaleheader).

Automated financial report generation (daily sales, monthly summaries, shop summaries, and invoice counts) with export utilities for Excel (.xlsx) and PDF formats.

Tech Stack

Frontend: React.js, Vite, Tailwind CSS, Axios, Lucide UI, xlsx, jspdf, jspdf-autotable.

Backend: Node.js, Express.js.

Database: MySQL (managed via mysql2).

Product Perspective

The system operates as a client-server web architecture. The React frontend handles user interactions, dynamic routing, and data presentation. The Express.js backend exposes RESTful API endpoints that process business logic, handle role-based filters, and interact with the MySQL database.

User Classes and Characteristics

Administrators (Admins): Have unrestricted global access to the system. They can register partner companies, modify operator profiles, inspect all transaction logs across the platform, and generate/download official financial reports.

Operating Environment

Client browsers running modern web standards (Chrome, Firefox, Edge, Safari).

Node.js runtime environment for the backend server.

MySQL relational database server instance.

System Features and Functional Requirements

Authentication Module (Login.jsx)

Description: Secures system entry for administrators.

Requirements:

The system shall provide an input form for username (email) and password.

Credentials shall be validated against the tbllogin database table via /api/login.

Successful authentication shall save user session data to browser localStorage and redirect the user to the dashboard.

Invalid credentials shall trigger a 401 Unauthorized response and display an error message (Username or password is wrong).

Company Directory & Management Module (Companies.jsx)

Description: Centralized management registry for partner operators (e.g., Uber, PickMe).

Requirements:

The system shall display partner companies using interactive directory tabs and a detailed data view.

The system shall track and display entity parameters including companyId, companyname, address, contact, email, fileCount, and lastDocCount.

Administrators shall be able to register new partners (POST /api/companies), update existing profiles (PUT /api/companies/:id), or delete outdated partner records (DELETE /api/companies/:id).

Sales & Transaction Dashboard (SalesDashboard.jsx)

Description: Real-time interface for tracking taxi trip logs and financial entries.

Requirements:

The system shall fetch and display transaction records from tblsaleheader joined with tblcompany.

The interface shall provide a manual data synchronization option via a refresh mechanism.

Reporting and Analytics Suite (Report.jsx)

Description: Financial analytics utility with document export features.

Requirements:

The system shall generate four primary report categories:

Daily Sales Report filtered by exact dates (YYYY-MM-DD).

Monthly Summary aggregated by month (YYYY-MM).

Company Monthly Summary filtered by a specific Operator ID and month.

Invoice Count Report tracking volume distribution.

The system shall provide one-click export tools to bundle active data tables into Excel (.xlsx) spreadsheets or formatted PDF documents via jsPDF.

4. Database Schema Requirements

tbllogin Table: Stores authorization credentials (userName, password).

tblcompany Table: Stores partner metadata (companyId [PK], companyname, address, contact, email, fileCount, lastDocCount).

tblsaleheader Table: Stores transaction headers (headerId [PK], OperatorId [FK], OperatorName, invoiceNumber, vehicleNo, vehicleModel, transactionDate, transactionTime, destination, paymentType, noOfVehicle, invoiceAmount, cancellation, TotalAmount, DateTime).

5. Non-Functional Requirements

Security: Password credentials and API inputs must be handled securely, with session data isolated per user profile.

Usability: The interface must use responsive card layouts, clean typography, and clear visual hierarchy built with Tailwind CSS.

Performance: Real-time data retrieval via Axios and MySQL queries should execute with minimal latency under normal operational volume.